Efficient Bayesian Inference for Multivariate Factor Stochastic Volatility Models

(Kastner et al. 2017)

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Outline

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From Univariate to Multivariate Stochastic Volatility

- **Recall Assignment Task 2:** We modelled daily CHF/USD returns with t_ν-errors and latent weights ω_i, giving each observation its own effective variance σ^2/ω_i — a *univariate* stochastic volatility (SV) model.
- **Natural extension:** With m assets we need not only time-varying *variances* but also time-varying *covariances*. The full covariance matrix has $\frac{m(m+1)}{2}$ free elements — growing **quadratically** in m.
- **The curse of dimensionality:** For $m = 26$ exchange rates this gives 351 covariance parameters *per time point*. Direct modelling is infeasible.
- **This paper's answer:** Impose a low-dimensional **factor structure** ($r \ll m$ latent factors) and let *each* factor follow its own SV process — parsimonious yet flexible.

Building Block 1: The Factor SV Model

Intuition: Decompose m-dimensional returns into a low-dimensional common component (factors) plus series-specific noise. Both components have *time-varying* volatility.

Observation & Factor Equations

$$\underbrace{\mathbf{y}_t}_{m\times 1} = \underbrace{\mathbf{\Lambda}}_{m\times r}\underbrace{\mathbf{f}_t}_{r\times 1} + \underbrace{\varepsilon_t}_{m\times 1}, \qquad \mathbf{f}_t \mid \mathbf{h}_t \sim \mathcal{N}_r(\mathbf{0}, V_t), \quad \varepsilon_t \mid \mathbf{h}_t \sim \mathcal{N}_m(\mathbf{0}, U_t)$$

- $\mathbf{\Lambda}$: $m \times r$ **factor loadings** — how much each asset responds to each factor
- $U_t = \mathrm{diag}(e^{h_{1t}}, \ldots, e^{h_{mt}})$: **idiosyncratic** time-varying variances
- $V_t = \mathrm{diag}(e^{h_{m+1,t}}, \ldots, e^{h_{m+r,t}})$: **factor** time-varying variances
- Each log-variance follows an independent **AR(1)**:
  $h_{it} = \mu_i + \phi_i(h_{i,t-1} - \mu_i) + \sigma_i \eta_{it}$

Key quantity: time-varying covariance matrix

$$\mathrm{cov}(\mathbf{y}_t \mid \mathbf{h}_t) \;=\; \mathbf{\Lambda} V_t \mathbf{\Lambda}' + U_t$$

Building Block 2: Identification

Problem: The factor model has a fund. **scaling ambiguity**. For any scalar $c \neq 0$:

$$\mathbf{\Lambda}\mathbf{f}_t = \underbrace{(c\,\mathbf{\Lambda})}_{\mathbf{\Lambda}^*}\underbrace{(c^{-1}\mathbf{f}_t)}_{\mathbf{f}_t^*}$$

The data cannot distinguish $(\mathbf{\Lambda}, \mathbf{f}_t)$ from $(c\mathbf{\Lambda}, c^{-1}\mathbf{f}_t)$ — the model is **not identified** without additional constraints.

Classical fix

Set $\lambda_{jj} = 1$ for each factor, treat the factor variance level μ_{m+j} as free.

- \+ Simple to implement
- − Imposes a series ordering
- − Creates artificial "leading" factors

Paper's approach

Fix $\mu_{m+j} = 0$ (factor variance averages to 1 over time), leave λ_{jj} *unrestricted*.

- \+ No ordering imposed
- \+ Leadership shared across series
- \+ Residual ambiguity in λ_{jj} becomes a **computational tool**

Why this matters for MCMC

Unresolved non-identifiability $\Rightarrow$ multimodal posterior $\Rightarrow$ MCMC chains get stuck.

Building Block 3: Priors & Data Augmentation

Priors (independently for each SV process $i = 1, \ldots, m + r$):

$$\mu_i \sim \mathcal{N}(b_\mu, B_\mu), \qquad \frac{\phi_i + 1}{2} \sim \mathcal{B}(a_0, b_0), \qquad \sigma_i^2 \sim B_\sigma \cdot \chi_1^2$$

For each unrestricted element of $\mathbf{\Lambda}$: $\lambda_{ij} \sim \mathcal{N}(0, B_\lambda)$ *(Ridge-type shrinkage of small loadings)*

Data augmentation: Introduce latent variables $\mathbf{h} = \{h_{i,\bullet}\}$ (log-volatility paths) and $\mathbf{f} = \{\mathbf{f}_t\}$ (factor paths) as "missing data".

Why augmentation helps — the key idea

- **Conditional on f and Λ:** each of the $m + r$ series reduces to an independent **univariate SV model** $\Rightarrow$ use `stochvol`.
- **Conditional on h:** $\mathbf{\Lambda}$ and $\mathbf{f}_t$ have **closed-form Normal posteriors** $\Rightarrow$ standard regression updates.

Direct parallel to Task 2: augmenting with ω_i turned an intractable t-likelihood into a conjugate Normal–Gamma system.

Building Block 4: The Standard Gibbs Sampler

Iterate the following three steps (Kastner et al. 2017, Algorithm 1):

Step (a) — Univariate SV updates ($m + r$ independent problems)

Conditional on $\mathbf{f}$ and $\mathbf{\Lambda}$, use the log-squared observation trick to write each series as a linear state-space model and update $(h_{i,\bullet}, \mu_i, \phi_i, \sigma_i)$ via `stochvol`.

Step (b) — Sample loadings $\mathbf{\Lambda}$

Conditional on $\mathbf{f}$ and $\mathbf{h}$, each row of $\mathbf{\Lambda}$ is a **weighted Normal regression** posterior — m independent problems, each with T observations, weighted by $e^{-h_{it}}$.

Step (c) — Sample factors $\mathbf{f}_t$

Conditional on $\mathbf{\Lambda}$ and $\mathbf{h}$, each $\mathbf{f}_t$ is a **multivariate Normal** posterior — T independent r-dimensional updates.

In principle correct — in practice broken. See next slide.

The Problem: Standard Gibbs Mixes Poorly

- **Root cause: $\mathbf{\Lambda}$** and $\mathbf{f}_t$ are **strongly correlated** in the joint posterior. If λ_{jj} is large, the factors f_{jt} must be small to keep Λf_t constant, and vice versa. Sequential updates cannot break this ridge-shaped dependence.
- **Consequence:** The chain inch-worms along a narrow ridge rather than freely exploring the parameter space.
- **Simulation evidence** ($m = 10$, $r = 2$, $T = 1000$, 100 datasets): **IF for λ_{11} exceeds 5000** on some datasets even after 5 million iterations — an effective sample size of only $\sim$1000 from 5 million draws.

Inefficiency Factor (IF) — reminder

$\mathrm{IF} = k$ means you need k correlated MCMC draws to match the accuracy of *one* independent draw.

IF = 5000 with $M = 100{,}000$ draws $\Rightarrow$ effective sample size of just 20. Practically useless for inference.

The ASIS Key Idea

Core insight (Yu and Meng 2011): Two *mathematically equivalent* parameterisations are often each efficient in *different* regions of the param. space. **Switching between them at every iteration** combines the best of both worlds.

Baseline param.

$$\mathbf{y}_t = \mathbf{\Lambda}\mathbf{f}_t + \varepsilon_t$$

λ_{jj} **free**, $\mu_{m+j} = 0$ fixed

Factor variance fixed at scale 1 on average.

Alternative param.

$$\mathbf{y}_t = \mathbf{\Lambda}^*\mathbf{f}_t^* + \varepsilon_t$$

$\lambda_{jj}^* = 1$ **fixed**, μ_{m+j} free

Scaling absorbed into the SV level.

The link (a simple change of variables):

$$\mathbf{f}_t^* = \lambda_{jj}\,\mathbf{f}_t, \qquad \mathbf{\Lambda}^* = \lambda_{jj}^{-1}\mathbf{\Lambda}$$

ASIS in one sentence

After each standard Gibbs step, transform to the alternative parameterisation, **resample** λ_{jj} there (where the posterior is better-behaved), then transform back. No extra model fitting — just a change of coordinates at each step.

Shallow vs. Deep Interweaving

Shallow interweaving

Transform: $\mathbf{f}_t^* = \lambda_{jj}\mathbf{f}_t$
Resample λ_{jj}^2 from a conjugate **GIG** posterior (see Appendix A1):

$$\lambda_{jj}^2 \mid \cdot \sim \mathrm{GIG}(p,\, a,\, b)$$

Transform back: $\mathbf{f}_t = (\lambda_{jj}^{\mathrm{new}})^{-1}\mathbf{f}_t^*$
Scope: $\mathbf{\Lambda}$ and $\mathbf{f}$ only.

Gain: $\approx$ 2–8$\times$

Deep interweaving

Transform: additionally shift the log-volatility path (see Appendix A2):

$$h_{m+j,t}^* = h_{m+j,t} + 2\log|\lambda_{jj}|$$

Now $\mu_{m+j} = \log\lambda_{jj}^2$ is free.
Resample λ_{jj} via **independence MH** using a Gaussian approximation to the conditional likelihood.

Scope: $\mathbf{\Lambda}$, $\mathbf{f}$ *and* $\mathbf{h}$.

Gain: up to **400**$\times$

Why deep beats shallow

Shallow leaves the log-volatility path $h_{m+j,\bullet}$ frozen, so the slow mixing between λ_{jj} and $h_{m+j,\bullet}$ persists. Deep interweaving breaks *both* dependencies simultaneously. **Neither strategy requires tuning.**

Simulation Evidence

Setup: $m = 10$, $r = 2$, $T = 1000$, 100 replications, 5.1M draws each.

	No interweaving	**Shallow**	**Deep**
λ_{11} (avg. IF)	> 1000	≈ 200	≈ 5
Median IF across $\mathbf{\Lambda}$	$\sim$400–5000	$\sim$50–400	$\sim$3–20
Efficiency gain	—	2–8$\times$	up to **400**$\times$
Factor $f_{1,T}$ (IF)	high	moderate	low
Factor log-var $h_{m+1,T}$ (IF)	high	high	substantially reduced

Source: Tables 1–2, (Kastner et al. 2017).

Table 1: Average inefficiency factors for $\mathbf{\Lambda}$.

- Trace plots: deep interweaving $\Rightarrow$ **immediate mixing**; standard sampler barely moves (Figure 1 of paper).
- IFs for standard sampler vary enormously (100–5000+) across datasets; deep interweaving reduces this **uniformly** to single digits.
- Log predictive Bayes factors (cf. Week 3) correctly recover $r = 2$.

Application: 26 EUR Exchange Rates (2005–2015)

Setup:

- $m = 26$ daily EUR rates, $T = 2649$ log-returns
- $r = 4$ factors selected via log predictive Bayes factors
- 500,000 post-burn-in draws
- Data used *as-is*: includes CHF spike (Jan. 2015) and RUB collapse (Dec. 2014)

Interpretable factors:

- **F1:** Global USD/CNY factor
- **F2:** Eastern Europe (ZAR, HUF, PLN; JPY *negative*)
- **F3:** Commodity currencies (AUD, NZD, CAD)
- **F4:** Tiger Cub economies (SGD)

Key findings:

- Factor volatilities spike around the **2008 crisis** and again in 2014–15
- CNY, HKD: near-perfect correlation with USD throughout
- RUB–USD correlation fell from ≈ 0.9 to ≈ 0.4 during 2008–09
- **Idiosyncratic SV** absorbs CHF and RUB outliers without distorting the factor structure — analogous to how ω_i absorbed outliers in Task 2

Takeaways & Practical Outlook

Connections to the lecture:

- **Data augmentation (Wk. 2–3):** Factor SV Gibbs is a direct generalisation of the ω_i augmentation in Task 2
- **Model selection (Wk. 3):** Bayes factors select r — same framework as lecture
- **Shrinkage (Wk. 3):** Gaussian prior on $\mathbf{\Lambda}$ acts as Ridge penalty
- **Key lesson:** A correct model with a poorly mixing sampler is useless; ASIS is a general remedy beyond this model

Practical applications:

- **Portfolio optimisation:** Replace sample covariance with posterior mean of Σ_t; integrate out estimation uncertainty via posterior draws for more stable Markowitz weights
- **Risk management:** Full predictive distribution of $\mathbf{y}_{T+1}$ $\Rightarrow$ exact Bayesian VaR/ES without Gaussian assumptions
- **Correlation trading:** Time-varying posterior correlations with uncertainty bands signal when stable relationships are breaking down
- `factorstochvol`

Bibliography

- Kastner, G., Frühwirth-Schnatter, S., & Lopes, H.F. (2017). Efficient Bayesian inference for multivariate factor stochastic volatility models. *Journal of Computational and Graphical Statistics*, 26(4), 905–917.
- Yu, Y., & Meng, X.-L. (2011). To center or not to center: An ancillarity-sufficiency interweaving strategy (ASIS) for boosting MCMC efficiency. *Journal of Computational and Graphical Statistics*, 20(3), 531–570.

A1: Why the GIG Posterior in Shallow Interweaving

Goal: derive $p(\lambda_{jj}^2 \mid \mathbf{f}^*, \mathbf{\Lambda}^*, \mathbf{h})$. Let $\psi = \lambda_{jj}^2$. Three components contribute:

(1) Likelihood from the scaled factors $f_{jt}^* \mid h_{m+j,t}, \psi \sim \mathcal{N}(0, \psi\, e^{h_{m+j,t}})$:

$$p(\mathbf{f}_j^* \mid \mathbf{h}, \psi) \;\propto\; \psi^{-T/2} \exp\left(-\frac{1}{2\psi} \sum_t \frac{(f_{jt}^*)^2}{e^{h_{m+j,t}}} \right)$$

(2) Induced prior on transformed loadings $\lambda_{ij}^* \mid \psi \sim \mathcal{N}(0, B_\lambda/\psi)$:

$$p(\boldsymbol{\lambda}_{-jj}^* \mid \psi) \;\propto\; \psi^{k_j/2} \exp\left(-\frac{\psi}{2B_\lambda} \|\boldsymbol{\lambda}_{-jj}^*\|^2 \right)$$

(3) Prior $\psi \sim \mathcal{G}(1/2,\, 1/(2B_\lambda))$: $\;\propto\; \psi^{-1/2} e^{-\psi/(2B_\lambda)}$

Combining all three gives a kernel of the form $\psi^{p-1} \exp\{-(a\psi + b/\psi)/2\}$, which is the **Generalised Inverse Gaussian** $\mathrm{GIG}(p, a, b)$ with:

$$p = \tfrac{1+k_j-T}{2}, \quad a = \tfrac{1}{B_\lambda}\left(1 + \|\boldsymbol{\lambda}_{-jj}^*\|^2\right), \quad b = \sum_t \tfrac{(f_{jt}^*)^2}{e^{h_{m+j,t}}}$$

Sampled efficiently via the `GIGrvg` R package.

A2: The Log-Volatility Shift in Deep Interweaving

Starting point: In the baseline param., $\mathrm{Var}(f_{jt}) = e^{h_{m+j,t}}$ with $\mu_{m+j} = 0$ fixed.

Step 1 — absorb λ_{jj} into the SV level. The scaled factor $f^*_{jt} = \lambda_{jj} f_{jt}$ has variance:

$$\mathrm{Var}(f^*_{jt}) = \lambda_{jj}^2\, e^{h_{m+j,t}} = \exp\Big(\underbrace{\log \lambda_{jj}^2 + h_{m+j,t}}_{=:\, h^*_{m+j,t}}\Big)$$

So the transformed log-variance is $h^*_{m+j,t} = h_{m+j,t} + 2\log|\lambda_{jj}|$, and the level of the new AR(1) process is $\mu_{m+j} = \log \lambda_{jj}^2$.

Step 2 — resample the level. The conditional likelihood $p(h^*_{m+j,1}, \ldots, h^*_{m+j,T} \mid \mu_{m+j}, \phi, \sigma^2)$ is Gaussian in μ_{m+j}, so a tight **independence MH** proposal is constructed analytically (matching mean and variance of this likelihood).

Step 3 — transform back. Given μ^{new}_{m+j}, set $\lambda^{\mathrm{new}}_{jj} = e^{\mu^{\mathrm{new}}_{m+j}/2}$ and recover:

$$h^{\mathrm{new}}_{m+j,t} = h^*_{m+j,t} - 2\log|\lambda^{\mathrm{new}}_{jj}|, \qquad \boldsymbol{\Lambda}^{\mathrm{new}} = \frac{\lambda^{\mathrm{new}}_{jj}}{\lambda^{\mathrm{old}}_{jj}} \boldsymbol{\Lambda}^{\mathrm{old}}$$